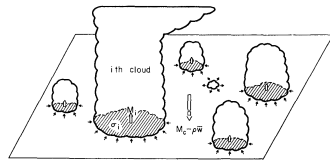


Step one of the theory: how a population of cumulus clouds changes the air around it.



$$\begin{aligned}\rho \partial_t \bar{s} &= D(\hat{s} - \bar{s} - L\hat{l}) + M_c \partial_z \bar{s} + LS_s \\ \rho \partial_t \bar{q} &= D(\hat{q}^* + \hat{l} - \bar{q}) + M_c \partial_z \bar{q} + LS_q\end{aligned}$$

detrainment: subsidence: large scale
cools, moistens warms, dries

Symbols

$s = c_p T + gz$ dry static energy, J kg^{-1}

q water-vapour mixing ratio, $\text{kg vapour} / \text{kg dry air}$

$\bar{s}, \bar{q}$ large-scale means, i.e. the environment

$M_c = \sum_i \rho \sigma_i w_i$ cloud mass flux, $\text{kg m}^{-2} \text{s}^{-1}$

σ_i, w_i area fraction, vertical velocity of cloud i

D mass detrained per unit volume and time

$\hat{\cdot}, *$ in the detrained air; at saturation

$\hat{l}$ liquid water in the detrained air

L, ρ latent heat of vaporization; air density

LS large-scale advection (+ radiation for s)

Clouds act on the environment only through M_c , D and $\hat{l}$

Arakawa–Schubert (1974): clouds and environment

Interaction of a Cumulus Cloud Ensemble with the Large-Scale Environment, Part I, Arakawa and Schubert (1974), J. Atmos. Sci. 31, 674–701. The theory predicts the collective effect of cumulus clouds and their response to a slowly evolving environment, including the subcloud mixed layer.

Cloud i covers the area fraction σ_i and carries the mass flux $\rho \sigma_i w_i$. The sum M_c drives compensating environmental subsidence. In a stably stratified atmosphere ($\partial_z \bar{s} > 0$) this warms, and since $\partial_z \bar{q} < 0$ it dries. Detrainment supplies cloud air and liquid water, whose evaporation cools ($-L\hat{l}$) and moistens. The latent heat released inside the clouds does not warm the environment directly: it keeps the clouds buoyant and so maintains M_c and the subsidence warming.

The variables are the dry static energy $s = c_p T + gz$, conserved by a dry parcel moving adiabatically, and the water-vapour mixing ratio q . Hats denote detraining-air values, the star denotes saturation, $\hat{l}$ is detrained liquid water, and L is latent heat. For legibility the slide groups the original large-scale terms as $LS_s = -\rho \bar{\mathbf{v}} \cdot \nabla \bar{s} - \rho \bar{w} \partial_z \bar{s} + \bar{Q}_R$ and $LS_q = -\rho \bar{\mathbf{v}} \cdot \nabla \bar{q} - \rho \bar{w} \partial_z \bar{q}$, with $\bar{Q}_R$ the radiative heating.

The equations assume a small active cloud fraction and a horizontally uniform environment. Figure 1 is the paper's conceptual mass-exchange sketch, not a measured horizontal cloud map. The parameterization problem is thus reduced to finding $M_c(z)$, $D(z)$ and $\hat{l}(z)$. Source: Sections 2–3, Figure 1, Equations (7), (12), (74)–(75).

Each cloud type is an entraining plume

2

The clouds are sorted into types by how fast they mix in outside air, the rate λ . Given the environment, each type's mass flux, energy and top follow.

$$\partial_z \eta = \lambda \eta \quad \Rightarrow \quad \eta = e^{\lambda(z-z_B)}$$

$$\eta h_c = h_M + \lambda \int_{z_B}^z \eta \bar{h} dz'$$

$$h_c(z_D) = \hat{h}^* \approx \bar{h}^* \quad (\text{top})$$

$$M_c(z) = \int_0^{\lambda_D(z)} \mathcal{M}_B(\lambda) \eta(z, \lambda) d\lambda$$

Symbols

λ fractional entrainment rate, km^{-1} : the type label. Large λ : small, diluted cloud with a low top (Fig. 8)

$\eta(z, \lambda)$ mass flux of type λ , normalized to 1 at z_B

$h = s + Lq$ moist static energy: conserved also when vapour condenses

h_c, h_M h inside the cloud; in the subcloud mixed layer

$\bar{h}^*$ h of the environment if it were saturated

z_B top of the mixed layer, where the clouds take their air

z_D cloud top, where buoyancy vanishes

$\mathcal{M}_B(\lambda)$ mass flux of type λ at z_B

$\lambda_D(z)$ type whose top is at height z

The environment fixes each plume: only the spectrum $\mathcal{M}_B(\lambda)$ is unknown

Paper review: Arakawa & Schubert (1974) | Slide 2

Each cloud type is an entraining plume

The ensemble is divided into entraining plumes with fractional entrainment rate λ , assumed constant with height. Entrainment means that outside air is mixed into the rising cloud. The normalized mass flux satisfies $\partial_z \eta = \lambda \eta$ and $\eta(z_B) = 1$, so it grows exponentially up to the plume top. The mass-flux reference level z_B is the mixed-layer top, which can lie below cloud base. Through the tower model of Simpson, λ is inversely proportional to the cloud radius: large λ means small clouds.

The moist static energy $h = s + Lq$ follows the entraining-plume budget. On the slide, the arguments are suppressed: $\eta(z, \lambda) h_c(z, \lambda) = h_M + \lambda \int_{z_B}^z \eta(z', \lambda) \bar{h}(z') dz'$. Entrainment dilutes mixed-layer air with environmental air, which has lower h in the mid-troposphere. The cloud top z_D is the level of vanishing buoyancy, $h_c(z_D) = \hat{h}^*(z_D)$, approximately $\bar{h}^*(z_D)$. Figure 8 shows the resulting cloud-top pressure for the reference tropical sounding. Follow the solid branch: increasing entrainment gives a lower top, hence higher pressure.

The integral for $M_c(z)$ sums the plume types still present at height z . Its upper limit $\lambda_D(z)$ selects types whose top reaches that level. The remaining unknown is the mass-flux density $\mathcal{M}_B(\lambda)$, not a number or radius density. Source: Section 4, Equations (77)–(97), Figure 8, and Section 5 for the mixed-layer boundary values.

$A(\lambda)$ is the kinetic energy that buoyancy gives clouds of type λ , per unit mass flux. The large scale builds it up; the clouds use it up.



$$A(\lambda) = \int_{z_B}^{z_D(\lambda)} \eta \underbrace{\frac{g(s_{vc} - \bar{s}_v)}{c_p \bar{T}}}_{\text{buoyancy}} dz$$

$$\frac{dA}{dt} = \int K(\lambda, \lambda') \mathcal{M}_B(\lambda') d\lambda' + F(\lambda) \simeq 0$$

$\simeq 0$: quasi-equilibrium, clouds adjust much faster than F changes.

Symbols

$s_v = s + c_p \bar{T}(0.608 q - l)$, virtual static energy: vapour makes air lighter, liquid water l heavier

$s_{vc}, \bar{s}_v$ in the cloud; in the environment

$g, \bar{T}$ gravity; environment temperature

$A > 0$ type λ can grow. For $\lambda = 0$, A is close to CAPE

$K(\lambda, \lambda')$ change of $A(\lambda)$ per unit $\mathcal{M}_B(\lambda')$; mostly negative

$F(\lambda)$ change of A by large-scale processes

Quasi-equilibrium turns $\mathcal{M}_B(\lambda)$ into the solution of an integral equation

Cloud work function and quasi-equilibrium

The cloud work function is the buoyancy integral weighted by normalized mass flux. It measures kinetic-energy generation per unit mass flux at the reference level: the kinetic-energy budget is $dK/dt = A\mathcal{M}_B - \mathcal{D}$. The environment determines A , including the mixed-layer contribution. Positive A permits buoyant energy production but does not by itself prescribe cloud counts or triggering. For non-entraining clouds, $\lambda = 0$, the integral is close to what is now called CAPE; entraining types also need a moist environment.

The map shows the feedback: large-scale forcing changes A , buoyancy supports cloud activity, and the ensemble modifies the environment and thus its own work function. Net stabilization dominates the adjustment. Individual entries of K need not all be negative, since detrainment by shallow clouds can help deeper clouds.

The tendency equation separates the cloud contribution, $\int_0^{\lambda_{\max}} K(\lambda, \lambda') \mathcal{M}_B(\lambda') d\lambda'$, from the large-scale forcing F : cooling by large-scale ascent, advection, radiation and surface fluxes. Under quasi-equilibrium the tendency is small compared with these two contributions, giving the integral equation that closes the mass-flux spectrum. The equality holds for active cloud types, with a nonnegative spectrum and an inequality for absent types. This is the principal theoretical result of Part I. It does not impose a steady temperature or moisture profile. Source: Sections 6–7, Equations (132)–(133), (142), (150) and (158). The map summarizes these equations.

Observations support quasi-equilibrium

4

Test with tropical data. If clouds did not respond, A would change as fast as the forcing (dashed line). The observed change of A stays near zero.

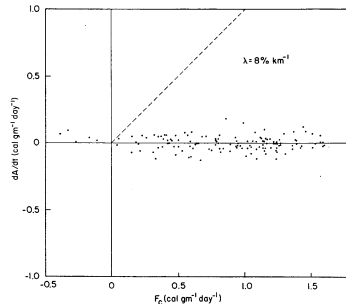


Fig. 13: observed dA/dt against F_C , for $\lambda = 8\% \text{ km}^{-1}$.

Two timescales

$$\tau_{\text{adj}} \sim 10^3\text{--}10^4 \text{ s}$$

15 min to 3 h

$$\tau_{\text{LS}} \gtrsim 10^5 \text{ s}$$

a day or more

Observed:

$$|dA/dt| \ll |F_C|$$

Symbols

F_C change of A that large-scale processes in the cloud layer would produce alone

dA/dt observed change of A

τ_{adj} time for the clouds to remove A

τ_{LS} time over which the large-scale forcing changes

cal g^{-1} energy per unit mass; 1

$\text{cal g}^{-1} \approx 4.2 \text{ kJ kg}^{-1}$

Clouds nearly cancel the forcing: A changes much less than F_C predicts

Observations support quasi-equilibrium

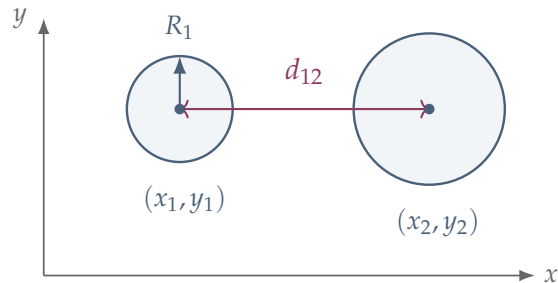
The paper estimates a cloud adjustment time of $10^3\text{--}10^4 \text{ s}$ and a large-scale forcing time of at least 10^5 s . This separation motivates dropping the relatively small work-function tendency in the closure equation. The estimates are orders of magnitude, not measured lifetimes of individual thermals.

Figure 13 compares the observed tendency of the cloud work function with cloud-layer forcing F_C from the Marshall Islands data of Yanai et al. (1973). The upper panel, reproduced here, uses $\lambda = 8\% \text{ km}^{-1}$. Without cloud adjustment, the observations would lie along the dashed line $dA/dt = F_C$. Instead, the tendency is generally much smaller than the forcing, which reaches about $1.5 \text{ cal g}^{-1} \text{ day}^{-1}$. The lower panel for $\lambda = 16\% \text{ km}^{-1}$ gives the same qualitative result.

The test supports quasi-equilibrium in this setting. It omits the mixed-layer forcing F_M , which could not be evaluated from the conventional observations. Figure 14 separately shows that temperature and moisture profiles evolve while their combined work function remains small. Numerical solution methods and applications of the closure are deferred to Part II. Source: Section 7, Equations (146)–(154), Figures 13–14, and Section 8.

Our question: how are thermals arranged in a horizontal plane? The theory sorts clouds by type, not by position.

Spatial quantities (schematic)



Sought, not in the paper

R_i radius of thermal i at a fixed height

(x_i, y_i) position of its centre

d_{ij} distance between thermals i and j

Why the theory does not answer it

A spectrum over λ has no positions or distances.

$\mathcal{M}_B$ mixes cloud number, lifetime and single-cloud mass flux, never separated.

The environment is assumed horizontally uniform.

A mass-flux spectrum leaves the horizontal thermal field unspecified

No 2D distribution of thermals

The target of our reading was the distribution of thermals in a horizontal plane. The sketch defines sizes, positions and centre-to-centre separations. It is illustrative only. Arakawa and Schubert do not report distributions of these thermal quantities.

The model resolves cloud types through their fractional entrainment and vertical structure, while closing their collective mass-flux contribution. A spectrum over λ is not a two-dimensional spatial distribution. The paper explicitly separates mass flux from cloud number, lifetime and single-cloud transport, which are not determined independently (Eq. 145). Its use of a relation between entrainment and cloud radius does not yield a measured thermal-radius distribution.

Horizontal uniformity of the environment is a modelling assumption, not evidence for randomly or uniformly positioned thermal centres. The cloud sketch in Figure 1 likewise supplies no spatial statistics. Source: Sections 4 and 7, particularly the discussion of Equations (94) and (145).